SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: Any 4 **Total Marks:40**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I E.Ragul / 192571032(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
E.Ragul

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.

Tasks:
a) Identify all actors involved in the system.
b) List the primary use cases.
c) Define relationships (include, extend, generalization).
d) Construct a complete Use Case Diagram representing system functionality.

(10 Marks)

2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.

Tasks:
a) Identify classes with attributes and methods.
b) Define relationships (association, aggregation, inheritance).
c) Specify multiplicity between classes.
d) Design a detailed Class Diagram for the system.

(10 Marks)

3. Sequence Diagram Problem

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.

Tasks:
a) Identify objects involved in the process.
b) Define message flow between objects.
c) Represent activation and lifelines.
d) Construct a Sequence Diagram showing complete interaction flow.

(10 Marks)

4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.

Tasks:
a) Identify all activities in the workflow.
b) Add decision nodes for approval/rejection.
c) Include parallel processing where applicable.
d) Design an optimized Activity Diagram.

Answers:

1. Use Case Diagram – Hospital Management System

a) Actors Involved

1. **Patient**
   - Book appointment
   - View appointment status
   - View medical records
2. **Doctor**
   - View appointments
   - Update medical records
   - Prescribe treatment
3. **Administrator**
   - Manage hospital resources
   - Manage doctors
   - Manage patient records
   - Generate reports

b) Primary Use Cases

- Register/Login
- Book Appointment
- View Appointment
- Cancel Appointment
- Update Medical Record
- View Medical Record
- Prescribe Treatment
- Manage Hospital Resources
- Manage Staff Information

c) Relationships

Include Relationship:

- **Book Appointment** → <<include>> Check Doctor Availability
- **Update Medical Record** → <<include>> View Patient Record
- **Generate Report** → <<include>> Collect Data

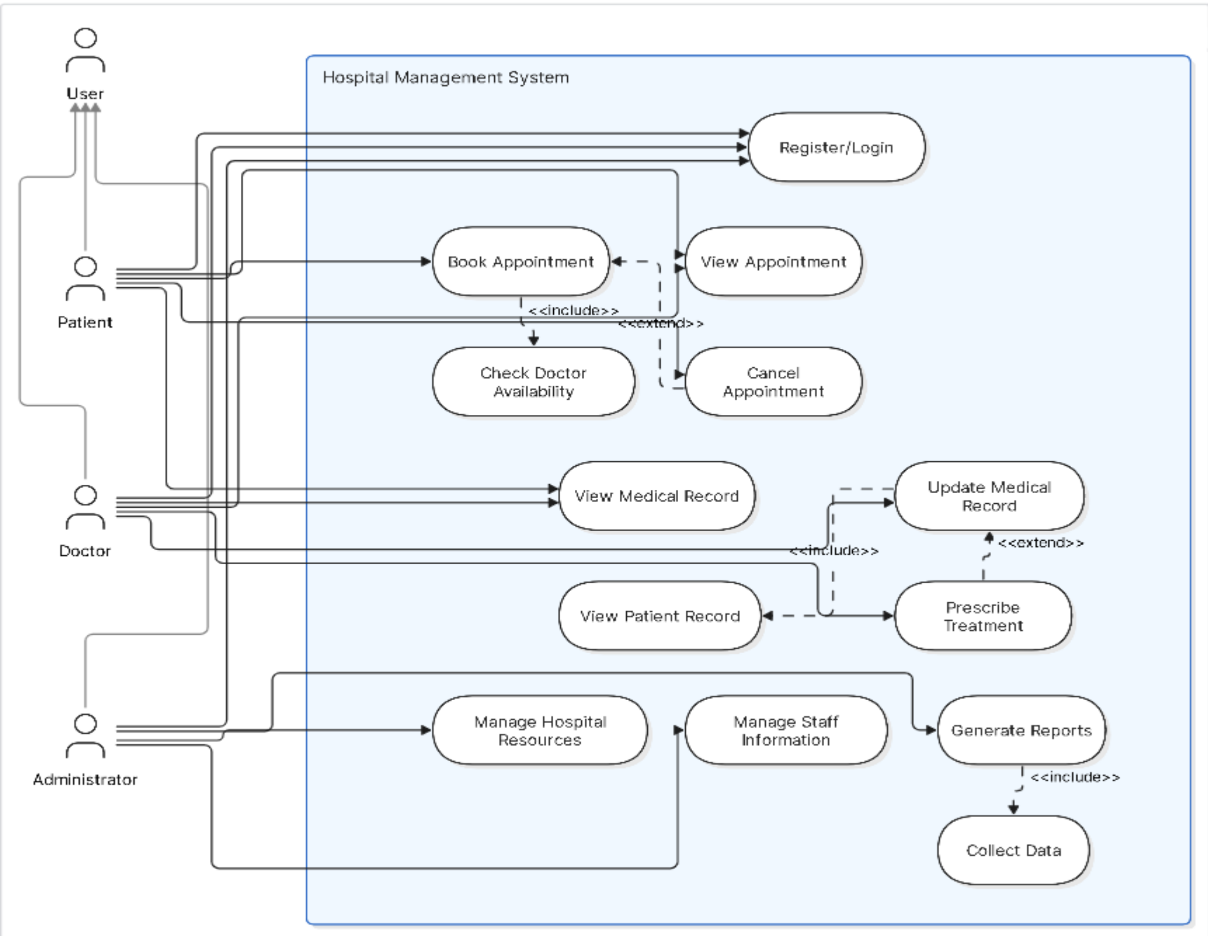
Extend Relationship:

- **Cancel Appointment** → <<extend>> Book Appointment
- **Prescribe Treatment** → <<extend>> Update Medical Record

Generalization:

- **User**
  - Patient
  - Doctor
  - Administrator

d) Use Case Diagram:



2. Class Diagram – Library Management System

a) Classes with Attributes and Methods

Book

Attributes

- bookId
- title
- author
- publisher
- status

Methods

- addBook()
- updateBook()
- searchBook()

Member

Attributes

- memberId
- name
- address
- phone

Methods

- borrowBook()
- returnBook()

Librarian

Attributes

- librarianId
- name
- salary

Methods

- issueBook()
- collectBook()
- manageBooks()

IssueRecord

Attributes

- issueId
- issueDate
- returnDate
- fineAmount

Methods

- calculateFine()
- updateRecord()

Person (Super Class)

Attributes

- id
- name

Methods

- login()
- logout()

b) Relationships

Inheritance

```
Person
  |
  |---- Member
  |
  |---- Librarian
```

Association

- Member borrows Book
- Librarian issues Book
- IssueRecord stores transaction details

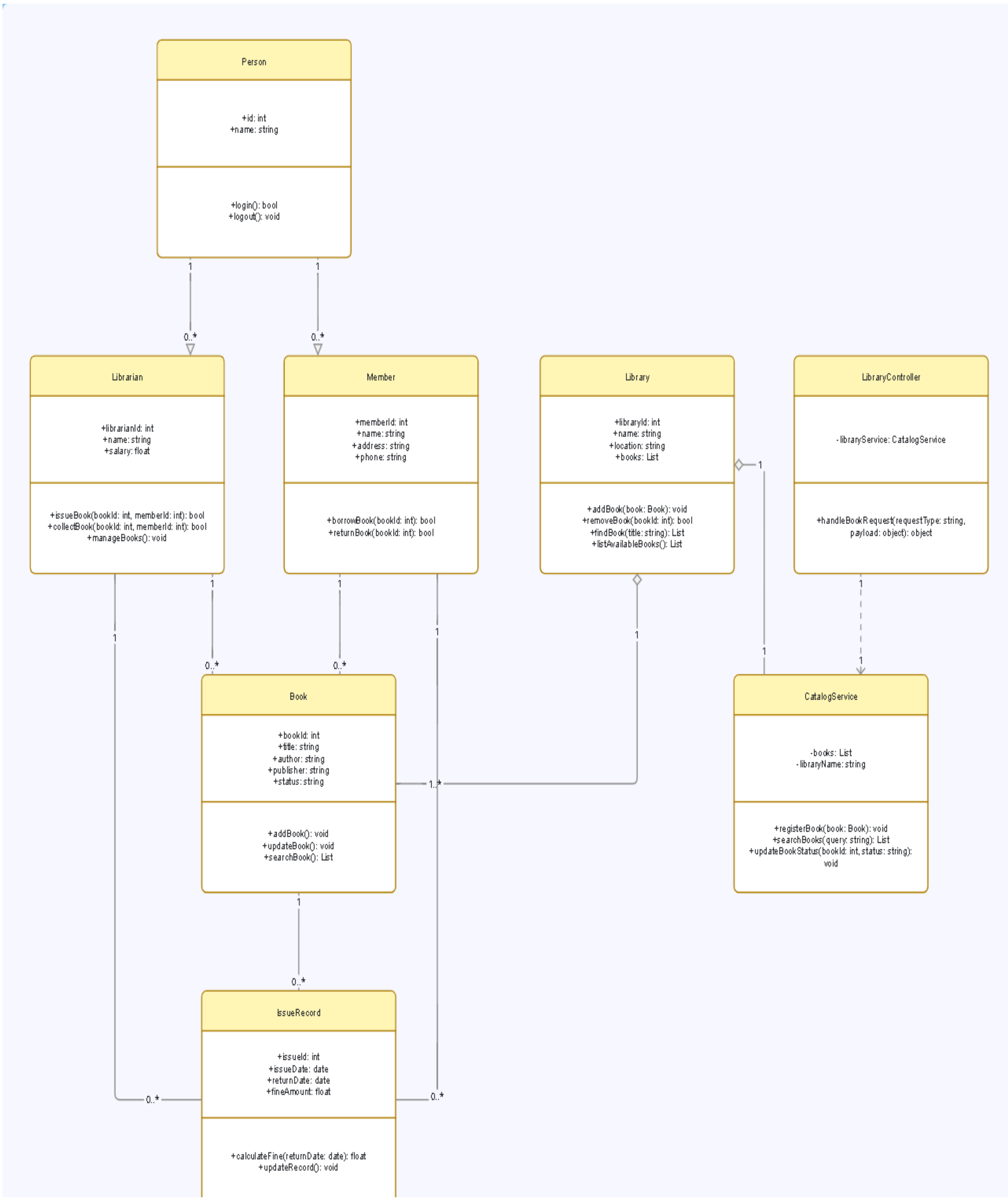
Aggregation

- Library contains many Books

c) Multiplicity

Relationship	Multiplicity
Member – IssueRecord	1 : *
Book – IssueRecord	1 : *
Librarian – IssueRecord	1 : *
Library – Book	1 : *

d) Class Diagram



3. Sequence Diagram – Online Food Ordering System

a) Objects Involved

1. Customer
2. Food Menu
3. Order System
4. Payment Gateway
5. Database
6. Notification Service

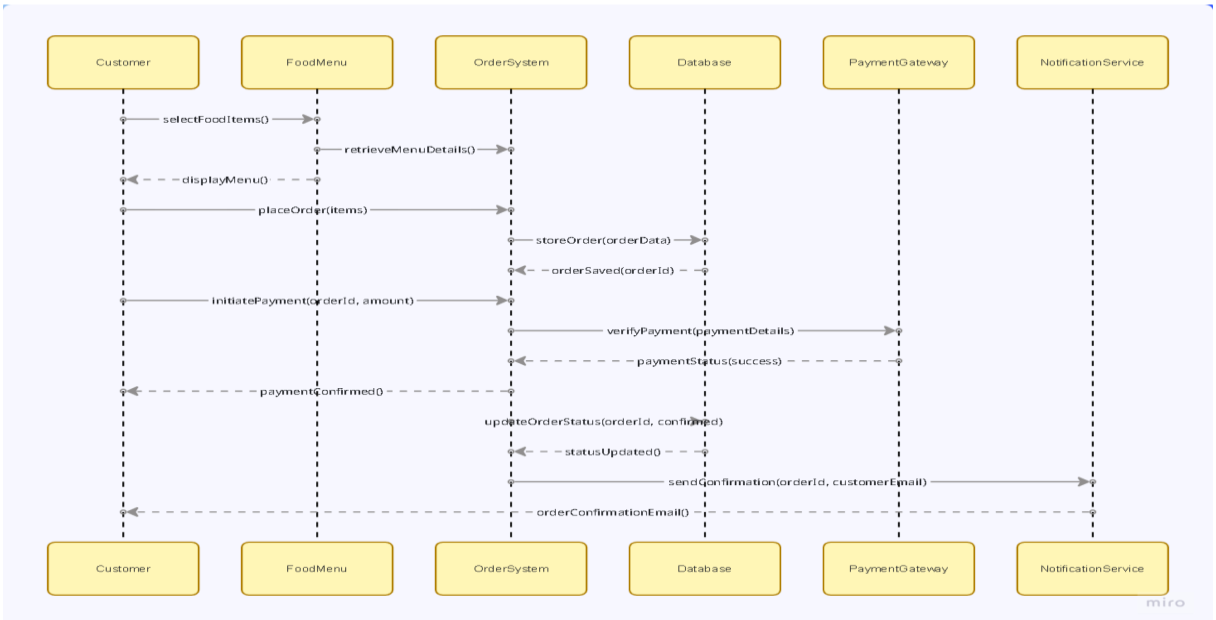
b) Message Flow

1. Customer selects food items.
2. Order System retrieves menu details.
3. Customer places order.
4. Order System stores order in database.
5. Customer initiates payment.
6. Payment Gateway verifies payment.
7. Payment status returned.
8. Order System confirms order.
9. Notification Service sends confirmation.

c) Lifelines and Activations

- Each object has a vertical lifeline.
- Activation bars appear when processing requests.
- Messages are represented by arrows between lifelines.

d) Sequence Diagram



4. Activity Diagram – Student Admission System

a) Activities

1. Start
2. Submit Application
3. Upload Documents
4. Verify Documents
5. Approve/Reject Application
6. Pay Admission Fee
7. Confirm Enrollment
8. End

b) Decision Nodes

Decision 1: Documents Valid

- Yes → Continue
- No → Reject Application

Decision 2: Fee Paid

- Yes → Enrollment Confirmation
- No → Payment Retry

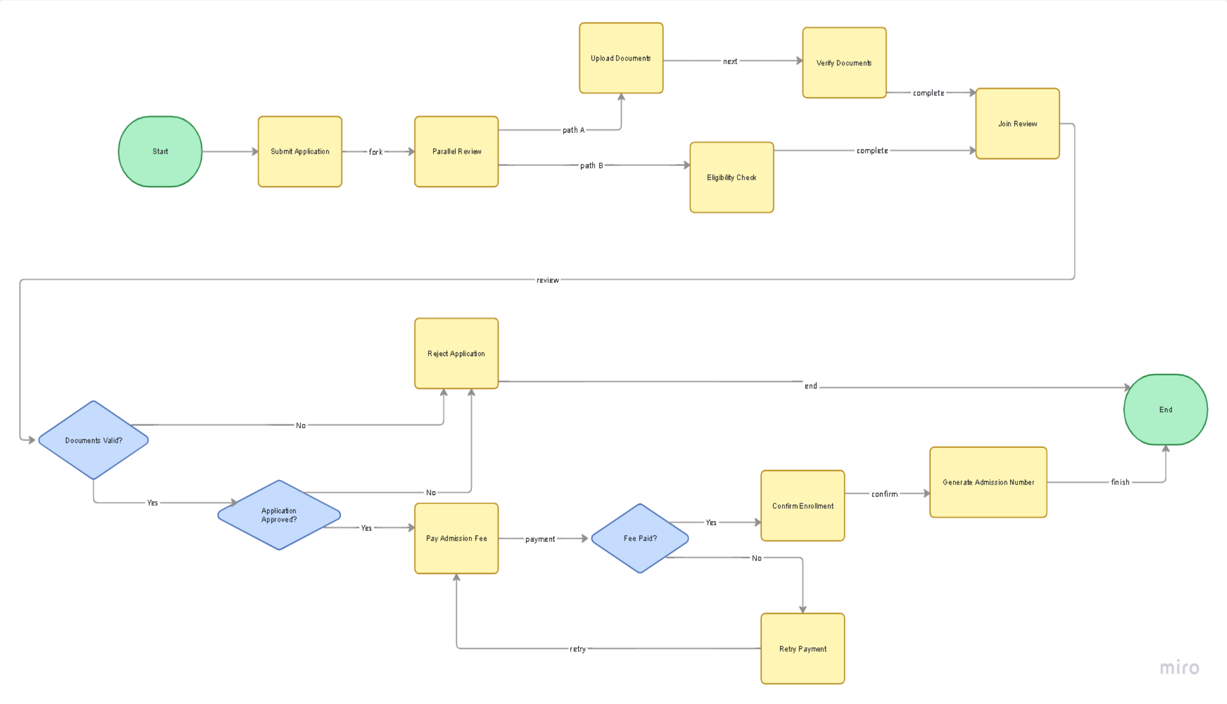
c) Parallel Processing

After application submission:

- Document Verification
- Eligibility Check

Both activities run simultaneously and then merge before approval.

d) Activity Diagram



RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation